Decomposition programme on ECC2K-130: plan report

Date: 2026-10-16. **Study:** `research/decomposition_programme_20261016/` (source: `PLAN.md`). **Status:** plan only. Nothing in this report is a measurement, and no work below 2^{61} is claimed.

Question

What is left for summation-polynomial decomposition on $y^2 + xy = x^3 + 1$ over $\mathbb{F}_{2^{131}}$, with $\#E = 4r$ and matched rho at $2^{60.8090}$?

Baseline and closed branches

The baseline is the linear (A, P) pair oracle over a geometric factor base (frozen, PR #1442). Its gap to rho is 42.47 bits, or 55.22 bits charged with linear algebra at $\omega = 2$.

These branches are closed by frozen experiments:

- higher-degree XL on the bilinear core (#1420);
- XL on the trilinear system with t unknown (#1429);
- the three-summand linearization (#1442);
- graph-cycle IC (#1510);
- torsion and model invariants, and additive energy (#1509).

Scoping facts (proposed; not cited)

These came from throwaway checks in the planning session. They are re-established in Rust under frozen protocols in W1–W3.

1. The order of 2 modulo 131 is 130. So $\mathbb{F}_{2^{131}}$ has no small Frobenius-invariant subspace, and the same holds at $n = 13, 19$. At $n = 23$ the order is 11, which gives an 11-dimensional invariant subspace and makes it a positive control.
2. $\dim(VV) \geq 2l - 1$ for every sampled V, and geometric V attains it. This is the linear Kneser bound (Hou–Leung–Xiang). If it holds, the pair-oracle reach $l \leq \frac{n+1}{3}$ is a ceiling for every subspace factor base.
3. Under the `gcost.py` heuristics, a one-shot $m = 4$ test fits the rho budget at about 2^{11} group operations per attempt when it handles same-conjugate decompositions only. It fits at about 2^{34} when it also handles mixed Frobenius conjugates. For geometric V, $\dim\left(V \cdot V^{2^k}\right)$ was 13–19 against the bound 9 at $n = 19, l = 5$.

Workstreams and gates

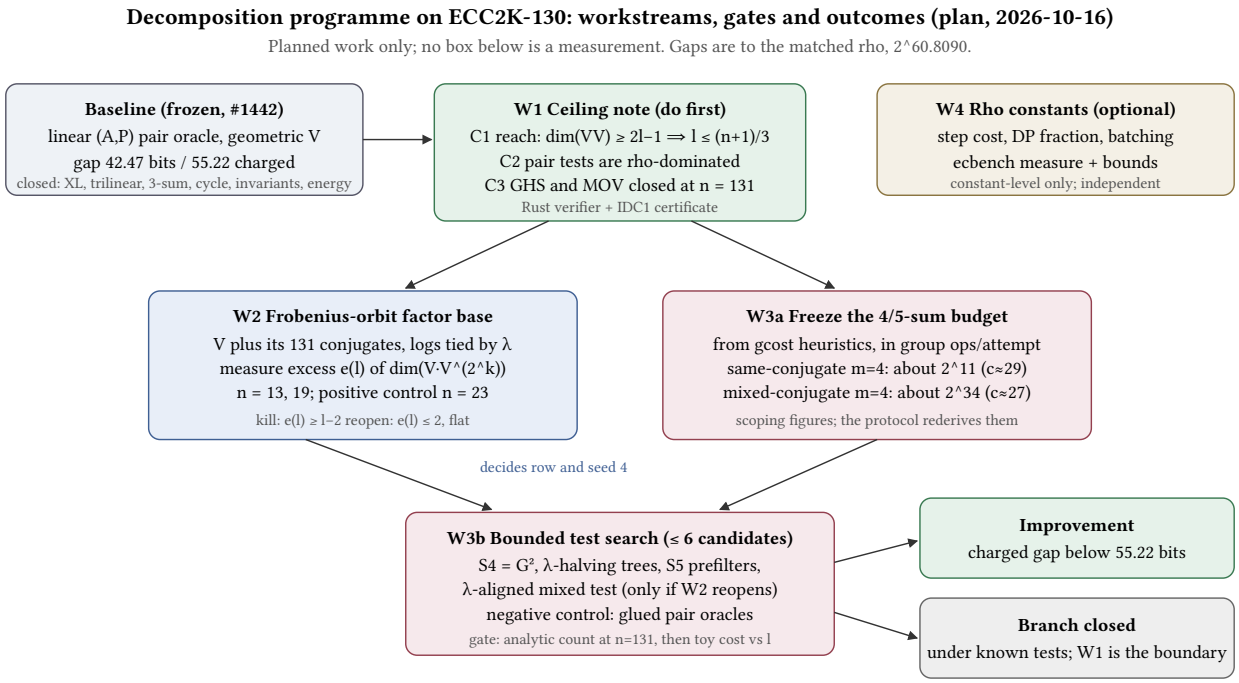


Figure 1: Planned workstreams, their order and their outcomes. Source: `workstreams.svg`, hand-written, in this directory. No box is a measurement.

- **W1, ceiling note.** Statements C1–C3, their proofs and scope, a Rust verifier for the $2l-1$ bound, and an identity certificate. Do this first.
- **W2, Frobenius-orbit factor base.** Measure the excess $e(l)$ of $\dim\left(V \cdot V^{2^k}\right)$ over $2l-1$. Kill if $e(l) \geq l-2$; reopen if $e(l) \leq 2$ with no trend in l.
- **W3, four- or five-summand test.** Freeze the budget first. Then run a bounded search over at most 6 candidates, with a glued-pair-oracle negative control and an analytic-then-toy gate.
- **W4, rho constants.** Optional and independent. Uses ecbench measure and bounds, and claims nothing beyond constants.

Success and kill rules

Any verified improvement over the baseline counts as a success, reported with its distance to rho (owner criterion, 2026-10-15). A branch closes only when no candidate can improve the baseline.

Graphs checked

No canonical graph changes, because this PR lands no measured value. No scoreboard, leaderboard or timeline entry is added or altered.